Direct Numerical Simulations Reveal Three Growth Regimes during Mixed-Phase Cloud Glaciation

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Key Points:

- Three growth regimes are observed during mixed-phase cloud glaciation caused by entrainment and mixing.
- A higher level of flow turbulence increases the evaporation of cloud droplets and thereby glaciation.
- A hexagonal column geometry promotes glaciation while a hexagonal plate geometry suppresses glaciation compared to spherical ice crystals.

Abstract

The glaciation of a mixed-phase cloud is investigated at the microscale by use of direct numerical simulations (DNS) and Lagrangian particle tracking. The dry air is entrained into the cloud volume and subsequent turbulent mixing is simulated. The effects of flow turbulence intensity, ice number density, and ice crystal geometry on glaciation are studied. It is found that high turbulence has an expediting effect on cloud droplet evaporation and glaciation. A hexagonal column shape promotes ice growth and glaciation compared to a sphere with the same volume. However, a hexagonal plate shape suppresses glaciation relative to spherical ice crystals. A higher ice number density enhances glaciation. For the simulated parameters, three growth regimes are visible as the glaciation fraction increases. They are characterized as evaporation-dominated, deposition-enhanced, and deposition-only.

1 Introduction

Mixed-phase clouds are found at temperatures between $-40°$ to $0°$ C (Korolev et al., 2017). In this temperature range, the saturation vapor pressure and supersaturation are higher with respect to water than they are with respect to ice at a given temperature. Thus, the air can be supersaturated with respect to ice but subsaturated with respect to water. That will cause ice crystals to grow but cloud droplets to shrink in a mixed-phase cloud. The increase of ice mass at the expense of liquid mass is known as the Wegener-Bergeron-Findeisen (WBF) process (Storelvmo & Tan, 2015) and is commonly referred to as glaciation. Mixed-phase clouds can fully glaciate or persist in a mixed-phase state (Morrison et al., 2012). The glaciation or persistence of mixed-phase clouds is a source of uncertainty in climate models and it needs to be investigated at the microscale.

Mixed-phase cloud was studied analytically by Korolev (2007) who suggested that cloud droplets and ice crystals grow simultaneously if supersaturation is positive over both ice and water. In that scenario, both cloud droplets and ice crystals compete for water vapor. But during the WBF process, cloud droplets serve to feed the depositional growth of ice crystals by providing water vapor through evaporation. The analytical investigation by Pinksly et al. (2014) suggested that the mass of mixed-phase cloud can increase or decrease during glaciation depending on the mass and number density of cloud droplets and ice crystals. Mixed-phase clouds have also been examined numerically. Wang et al. (2024) studied the glaciation process using a large eddy simulation (LES) model. The authors reported that the rate of glaciation is higher if mixed-phase clouds grow in a homogeneous environment. Yang et al. (2015) applied LES to examine the growth of ice crystals in a mixed-phase cloud and found that a fraction of ice crystals gets trapped in turbulent eddies and grow or sublimate faster depending on the local supersaturation over ice. LES was used by Ovchinnikov et al. (2014) who pointed out that the ice number density affcets mixed-phase cloud growth and the partitioning between cloud droplets and ice crystals needs accurate representation in numerical models. Chen et al. (2023) investigated the evolution of mixed-phase clouds using direct numerical simulations (DNS) and suggested that the size distributions of cloud droplets are more sensitive to turbulence than that of ice crystals and that the initial liquid mass has to be large enough for mixed-phase condition to persist.

The geometry of ice crystals is important. Using a theoretical approach, Chen & Lamb (1994) showed that the aspect ratio of ice crystal changes as the ice mass grows with time. Bailey & Hallett (2004) experimentally demonstrated that the complexity of ice crystal shape increases if the supersaturation with respect to ice is higher. Sheridan et al. (2009) suggested that ice growth is isotropic if ice crystals are larger but is anisotropic for smaller ice crystals. Sulia & Harrington (2011) found that glaciation can be under-predicted by nearly an order of magnitude if ice crystals are assumed to be spherical instead of the precise shape. Hence, further investigation is needed to determine the effects of ice crystal shape on glaciation.

In the present study, we implement DNS with Lagrangian tracking of cloud droplets and ice crystals to study the glaciation of mixed-phase clouds. The objective is to better understand the interaction between turbulence, cloud droplets, and ice crystals. We modify our previous particle-based DNS model (Sharfuddin et al., 2026) to solve the ice phase equations. There are two novel aspects of this study. First, we simulate the entrainment of dry air into the mixed-phase cloud and subsequent turbulent mixing with variation of flow turbulence intensity and ice number density. Second, we study the depositional growth of ice crystals for three geometries: sphere, hexagonal column, and hexagonal plate, in a turbulent environment.

2 Governing Equations

The turbulent flow of air is governed by the incompressible Navier-Stokes equations:

$$\frac{\partial \boldsymbol{u}}{\partial t} + (\boldsymbol{u}.\nabla)\boldsymbol{u} = -\frac{1}{\rho_a}\nabla p + \nu\nabla^2\boldsymbol{u} + \boldsymbol{F}_b + \boldsymbol{F}_c \,, \tag{1}$$

$$\nabla.\boldsymbol{u} = 0 \,. \tag{2}$$

Here, $\boldsymbol{u}$ is the instantaneous air velocity vector, ν is the kinematic viscosity of air, p is the instantaneous air pressure, and ρ_a is the air density. $\boldsymbol{F}_c$ is an external force that maintains flow turbulence and is constructed in the Fourier space. $\boldsymbol{F}_b$ is the buoyancy force. The definitions of $\boldsymbol{F}_c$ and $\boldsymbol{F}_b$ can be found in Sharfuddin et al. (2026). The conservation equations for the temperature (T) and water vapor mixing ratio (q_v) are:

$$\frac{\partial T}{\partial t} + (\boldsymbol{u}.\nabla)T = \frac{L_w}{c_p}C_w + \frac{L_{ice}}{c_p}C_{ice} + \mu_T\nabla^2 T \,, \tag{3}$$

$$\frac{\partial q_v}{\partial t} + (\boldsymbol{u}.\nabla)q_v = -(C_w + C_{ice}) + \mu_v\nabla^2 q_v \,. \tag{4}$$

Above, L_w and L_{ice} are specific latent heats of condensation and deposition, respectively. C_w is the time rate of exchange between water vapor and liquid water while C_{ice} is the same between water vapor and solid ice. c_p is the specific heat of air and μ_T and μ_v are the molecular diffusivities of temperature and vapor, respectively. The environmental supersaturation (S_e) depends on the T and q_v, and is expressed as:

$$S_e = \frac{q_v}{q_{v,s}} - 1 = \frac{q_v}{621.97\,\frac{p_{sat}}{p - p_{sat}}} - 1 \,, \tag{5}$$

where $q_{v,s}$ is the saturation vapor mixing ratio and p_{sat} is the saturation vapor pressure which can be calculated with respect to water and ice (Huang, 2018) as:

$$p_{sat} = \begin{cases} p_{sat,w} = 611.2 \times \exp\left(\dfrac{17.67(T - 273.15)}{T - 29.65}\right) \\ p_{sat,ice} = 611.2 \times \exp\left(\dfrac{22.46(T - 273.15)}{T - 0.53}\right) \end{cases} . \tag{6}$$

The cloud droplets and ice crystals are tracked individually in Lagrangian fashion. The equations for the Lagrangian position ($\boldsymbol{X}$) and velocity ($\boldsymbol{V}$) vectors are written as:

$$\frac{d\boldsymbol{X}_i(t)}{dt} = \boldsymbol{V}_i(t) \,, \quad \frac{d\boldsymbol{V}_i(t)}{dt} = \frac{1}{(\tau_p)_i}[\boldsymbol{u}_i(\boldsymbol{X},t) - \boldsymbol{V}_i(t)] + \boldsymbol{g} \,, \quad (\tau_p)_i = \frac{2\rho_i(r_i)^2}{9\rho_a\nu} \,. \tag{7}$$

where the subscript 'i' denotes the i-th particle (cloud droplet or ice crystal). $\boldsymbol{u}_i$ is the instantaneous air velocity vector interpolated at the position of particle i, $\boldsymbol{g}$ is the gravity vector, τ_p is the particle response time, and ρ_i is the density of i-th particle. Conveniently, S_e and particle radius (r) can be written as:

$$S_e = \begin{cases} S_{e,w} & if \quad p_{sat} = p_{sat,w} \\ S_{e,ice} & if \quad p_{sat} = p_{sat,ice} \end{cases}, \quad r = \begin{cases} r_w & if \quad S_e = S_{e,w} \\ r_{ice} & if \quad S_e = S_{e,ice} \end{cases}. \tag{8}$$

The growth equations for cloud droplets and ice crystals is written as:

$$\frac{d\big(r_w(t)\big)_i}{dt} = \left(\frac{G_w}{r_w(t)} S_{e,w}(\boldsymbol{X},t)\right)_i, \quad \frac{d\big(r_{ice}(t)\big)_i}{dt} = \left(\frac{G_{ice}}{r_{ice}(t)} S_{e,ice}(\boldsymbol{X},t)\right)_i. \tag{9}$$

The growth factors G_w and G_{ice} are expressed as:

$$G_w = \frac{1}{\frac{L_w \rho_w}{k_T T}\left(\frac{L_w}{R_v T} - 1\right) + \frac{\rho_w R_v T}{\mu_v p_{sat,w}}}, \quad G_{ice} = \frac{1}{\frac{L_{ice} \rho_{ice}}{k_T T}\left(\frac{L_{ice}}{R_v T} - 1\right) + \frac{\rho_{ice} R_v T}{\mu_v f p_{sat,ice}}}. \tag{10}$$

where k_T is the thermal conductivity in air, R_v is the specific gas constant for water vapor, and f is the ice crystal shape factor. The f is unity for spherical ice crystals. For hexagonal ice crystals, the f is found from: (Westbrook et al., 2008)

$$f = 0.58(1 + 0.95\mathrm{A}^{0.75})(b/r_{ice}), \quad \mathrm{A} = \frac{h}{2b}. \tag{11}$$

Above, b is the side length and h is the height of a regular hexagonal prism, and A is the aspect ratio. The shape is hexagonal column if $\mathrm{A} > 1$ and hexagonal plate if $\mathrm{A} < 1$. r_{ice} is the equivalent radius of hexagonal ice crystals and is related to b and h by:

$$\frac{4}{3}\pi r_{ice}^3 = \frac{3\sqrt{3}}{2} b^2 h, \quad r_{ice} = 0.8527(b^2 h)^{1/3}, \quad b = 0.93\left(\frac{r_{ice}}{\mathrm{A}^{1/3}}\right). \tag{12}$$

Our model neglects ice nucleation, accretion, aggregation, and ventilation. It focuses on diffusion-limited ice growth only. With the foregoing, C_w and C_{ice} are determined as:

$$C_w(\boldsymbol{x},t) = \frac{4\pi\rho_w}{\rho_a a^3} \sum_{i=1}^{n_w} \Big(G_w r_w(t) S_{e,w}(\boldsymbol{X},t)\Big)_i, \tag{13}$$

$$C_{ice}(\boldsymbol{x},t) = \frac{4\pi\rho_{ice}}{\rho_a a^3} \sum_{i=1}^{n_{ice}} \Big(G_{ice} r_{ice}(t) S_{e,ice}(\boldsymbol{X},t)\Big)_i. \tag{14}$$

where ρ_w and ρ_{ice} are the densities of liquid water and solid ice, respectively. n_w is the number of cloud droplets in a grid cell having a volume of a^3 and n_{ice} is the same for ice crystals.

3 Numerical Details

Flow turbulence is assumed to be homogeneous and isotropic. Two turbulent kinetic energy (TKE) dissipation rates are generated by following the procedure described in Sharfuddin et al. (2026). They are denoted as 'low (L)' and 'high (H)', and correspond to the Kolmogorov length scales of 0.00092 m and 0.00226 m, respectively.

The physical model is that of a cubic domain inside which turbulence, cloud droplets, and ice crystals interact. The domain volume is 0.512^3 m^3, with triply periodic boundary conditions. At the start, the domain is divided equally into cloudy and clear-air regions. Cloud droplets and ice crystals are randomly placed in the cloudy region. A slab-like configuration is assumed so that the initial temperature and water vapor mixing ratio change sharply at the cloud/clear-air interface. This setup allows simulation of entrainment and mixing (Gao et al., 2018). The initial vapor mixing ratio is specified as:

$$q_v(z,t=0)=\begin{cases} q_v^{max}, & d-L\times 0.5\times 0.5\le z<d+L\times 0.5\times 0.5 \\ q_{v,e}, & elsewhere \end{cases}. \tag{15}$$

Above, $q_v^{max}=3.12\ g/kg$, $q_{v,e}=2.50\ g/kg$, and $d\equiv L/2$ is the width of the cloud slab. q_v^{max} corresponds to the cloudy region and $q_{v,e}$ represents the dry (subsaturated) region. The temperature field is initialized as:

$$T(t=0)=\langle T(t=0)\rangle-0.608\langle T(t=0)\rangle(q_v(z,t=0)-\langle q_v(t=0)\rangle)\,. \tag{16}$$

We simulate five cases: L-S-L, L-C-L, L-P-L, H-S-L, and H-S-H. The details are provided in Table I along with the values of relevant parameters. The ice crystal shape is hexagonal column for Case H-H-C, hexagonal plate for Case H-H-P, and spherical otherwise. The initial volume of an ice crystal is approximately 4.18 μm^3 for all cases. For a hexagonal column, the initial values of f, b, and h are 1.148, 0.685 μm, and 3.42 μm, respectively. The corresponding values for a hexagonal plate are 1.082, 1.26 μm, and 1.01 μm. The domain-average temperature is ~264 K initially. The common ice crystal shape at this temperature is hexagonal plate (Bailey & Hallett, 2009) but hexagonal column shape is possible depending on the $S_{e,ice}$. The assumption of spherical ice crystals is rather idealistic but necessary for comparison. The cloud droplet number density is 59.6 cm^{-3} and is much larger than the ice number density (Korolev et al., 2017). The number of grid points is 256^3 and the grid resolution is 2 mm for all cases. It is assumed that the aspect ratio of hexagonal ice crystals do not change with time because temperature variation is negligible in a microscale DNS model with a short simulation time.

Table 1

Details of the test cases

Case	TKE dissipation rate (m^2/s^3)	Number density of ice crystals (cm^{-3})	Initial radii of a cloud droplet, ice crystal (μm)	Initial surface area of an ice crystal (μm^2)	Aspect ratio (A) of an ice crystal
L-S-L	0.00068	0.011	8, 1	12.566	N/A
L-C-L	0.00068	0.011	8, 1	16.494	2.5
L-P-L	0.00068	0.011	8, 1	15.885	0.4
H-S-L	0.00816	0.011	8, 1	12.566	N/A
H-S-H	0.00816	0.055	8, 1	12.566	N/A

Case	Comments
L-S-L	low (L) turbulence, spherical (S) ice crystals, low (L) ice number density
L-C-L	low (L) turbulence, hexagonal column (C) ice crystals, low (L) ice number density
L-P-L	low (L) turbulence, hexagonal plate (P) ice crystals, low (L) ice number density
H-S-L	high (H) turbulence, spherical (S) ice crystals, low (L) ice number density
H-S-H	high (H) turbulence, spherical (S) ice crystals, high (H) ice number density

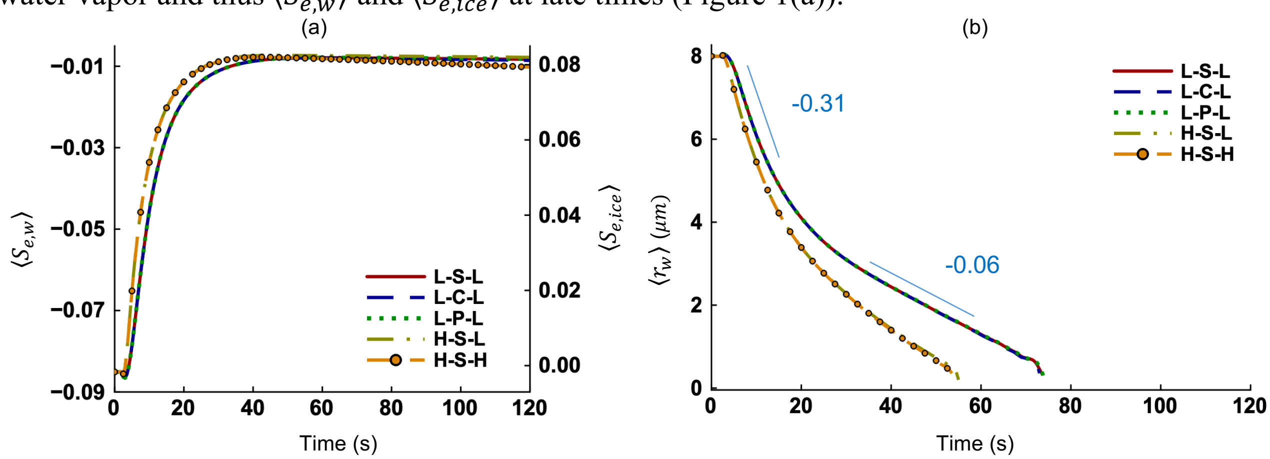
Parameters and their values

Symbol	Value	Unit	Symbol	Value	Unit
L_w	2.53×10^6	J/kg	$\mu_T = \mu_v$	2.9×10^{-5}	m^2/s
L_{ice}	2.84×10^6	J/kg	c_p	1005.0	J/kg/K
ρ_w	1000	kg/m^3	ρ_{ice}	917	kg/m^3
$p(t = 0)$	64455	N/m^2	$\langle T(t = 0)\rangle$	264.38	K
$p_{sat,w}$	315.73	N/m^2	$p_{sat,ice}$	289.60	N/m^2
ρ_a	0.85	kg/m^3	ν	1.96×10^{-5}	m^2/s
R_v	461.5	J/kg/K	k_T	0.0238	W/m/K

4 Results and Discussion

The mean supersaturation is negative with respect to water but positive with respect to ice at the initial time. Hence, ice crystals grow but cloud droplets evaporate, and the mixed-phase cloud glaciates. The effects of flow turbulence, ice number density, and ice crystal geometry on the thermodynamics and microphysics of glaciation are discussed.

The temporal evolution of the mean supersaturation fields is shown in Figure 1(a). Liquid water is converted into water vapor during the evaporation of cloud droplets and ice crystals grow by consuming that water vapor. Since the number density and mass of cloud droplets exceed that of ice crystals in mixed-phase clouds, more water vapor is produced by evaporation than consumed by ice crystals. Correspondong to the net increase of water vapor, the $\langle S_{e,w}\rangle$ and $\langle S_{e,ice}\rangle$ increase over time as shown in Figure 1(a). We see that the increase is faster when the turbulence level is high (Cases H-S-L and H-S-H). In contrast, the DNS study of Chen et al. (2023) found no impact of flow turbulence on the mean supersaturation fields because it did not account for entrainment and mixing like our model does. It is evident that the mean supersaturation profiles are identical for Cases L-S-L, L-C-L, and L-P-L. Thus, the ice crystal geometry has no influence on the $\langle S_{e,w}\rangle$ and $\langle S_{e,ice}\rangle$. A higher ice numer density (Case H-S-H) translates into larger ice crystal surface area. This allows more water vapor molecules to deposit on ice crystal surfaces, reducing the amount of water vapor and thus $\langle S_{e,w}\rangle$ and $\langle S_{e,ice}\rangle$ at late times (Figure 1(a)).



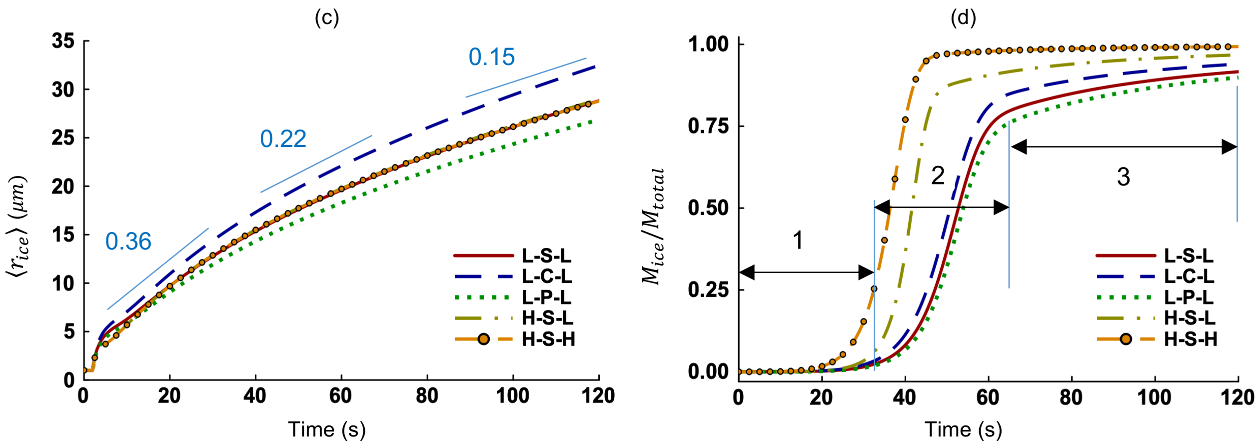


Figure 1. Time evolution of the (a) mean environmental supersaturations with respect to water ($\langle S_{e,w} \rangle$) and ice ($\langle S_{e,ice} \rangle$), (b) mean radius of cloud droplets ($\langle r_w \rangle$), (c) mean radius of ice crystals ($\langle r_{ice} \rangle$), and (d) ice mass or glaciation fraction. Three regimes are identified in Figure 1(d) for Case L-P-L. The plots for Cases H-S-L and H-S-H overlap in Figures 1(b) and 1(c). Also, the plots for Cases L-S-L, L-C-L, and L-P-L overlap in Figures 1(a) and 1(b).

The mean cloud droplet radius ($\langle r_w \rangle$) decreases faster (Figure 1(b)) and thus evaporation is stronger if the turbulence intensity is high (Cases H-S-L and H-S-H). Flow turbulence accelerates entrainment and mixing, which leads to a faster evaporation of cloud droplets (Hoffmann, 2020). Evaporation is quicker between 0 s to 20 s for all five cases. Evaporation moderates after 20 s and is completed by 55 s for high turbulence cases (H-S-L and H-S-H) and by 75 s for low turbulence cases (L-S-L, L-C-L, and L-P-L). For example, the slope of evaporative decay is -0.31 for Case L-C-L at early times and is -0.06 afterwards. The faster evaporation rate at early times corresponds to a larger negative supersaturation (Figure 1(a)). The domain becomes less subsaturated with time as evaporation progresses. We also report that ice crystal geometry has no impact on cloud droplet evaporation since the profiles are indetical for Cases L-S-L, L-C-L, and L-P-L in Figure 1(b). The mean radius of ice crystals ($\langle r_{ice} \rangle$) increases during deposition for all five cases (Figure 1(c)). The rate of increase slows down with time as indicated by the slopes (0.36 > 0.22 > 0.15). This happens because particle radius is inversely proportional to its growth rate (Equation (9)). The profiles of $\langle r_{ice} \rangle$ are identical for Cases L-S-L and H-S-L, implying that turbulence has negligible impact on ice crystal deposition. Since the number density of ice crystals is very low in mixed-phase clouds, turbulent mixing does not influence the deposition process. But the ice crystal geometry affects the evolution of $\langle r_{ice} \rangle$ considerably. We see in Figure 1(c) that spherical ice crystals (L-S-L) grow slower than hexagonal column ice crystals (L-C-L) but faster than those with hexagonal plate geometry (L-P-L). Seo et al. (2012) and Shi et al. (2025) used molecular dynamics to simulate ice crystal growth at ~260 K and found that the basal faces of hexagonal ice crystals grow slower than the prism faces. The authors explained that the basal faces grow layer by layer and vapor molecules need to complete a full rearrangement before the next layer can grow. The growth of prism faces follows a collective three-dimensional mechanism, which can incorporate vapor molecules readily. Since the basal faces constitute a larger proportion of the total surface area for a hexagonal plate than they do for a hexagonal column, ice growth is slower for the former as shown by our model. Note that Equation (11) was derived by Westbrook et al. (2008) using a Monte Carlo approach and it accounts for the attachment of vapor molecules to ice crystal surface. The agreement between our results and that of Seo et al. (2012), and Shi et al. (2025) is therefore not surprising.

The ratio of ice crystal mass to total cloud mass is denoted by M_{ice}/M_{total}. It indicates the fraction of the mixed-phase cloud that has been glaciated. Three regimes can be differentiated for all cases in Figure 1(d). Regime (1) is evaporation-dominated during which the rate of evaporation is highest. The mass of ice crystals, although growing fast, is still small compared to the total cloud mass and thus the glaciation fraction is negligible. As evaporation and deposition continue, the contribution of M_{ice} to M_{total} increases, making Regime (2) deposition-enhanced. We find that the glaciation fraction rises sharply during this period. By the end of Regime (2), the evaporation of cloud droplets is almost complete. The glaciation fraction keeps increasing during Regime (3), driven by the depostional growth of ice crystals. Thus, Regime (3) is characterized as deposition-only (Figure 1(d)). The glaciation rate is noticeably higher if the turbulence intensity is high (Cases H-S-L and H-S-H) compared to when it is low (Cases L-S-L, L-C-L, and L-P-L). With more ice crystals present, the glaciation rate is highest (H-S-H). For the same turbulence level, a hexagonal column geometry (L-C-L) enhances glaciation but a hexagonal plate geometry (L-P-L) reduces glaciation compared to spherical ice crystals (L-S-L).

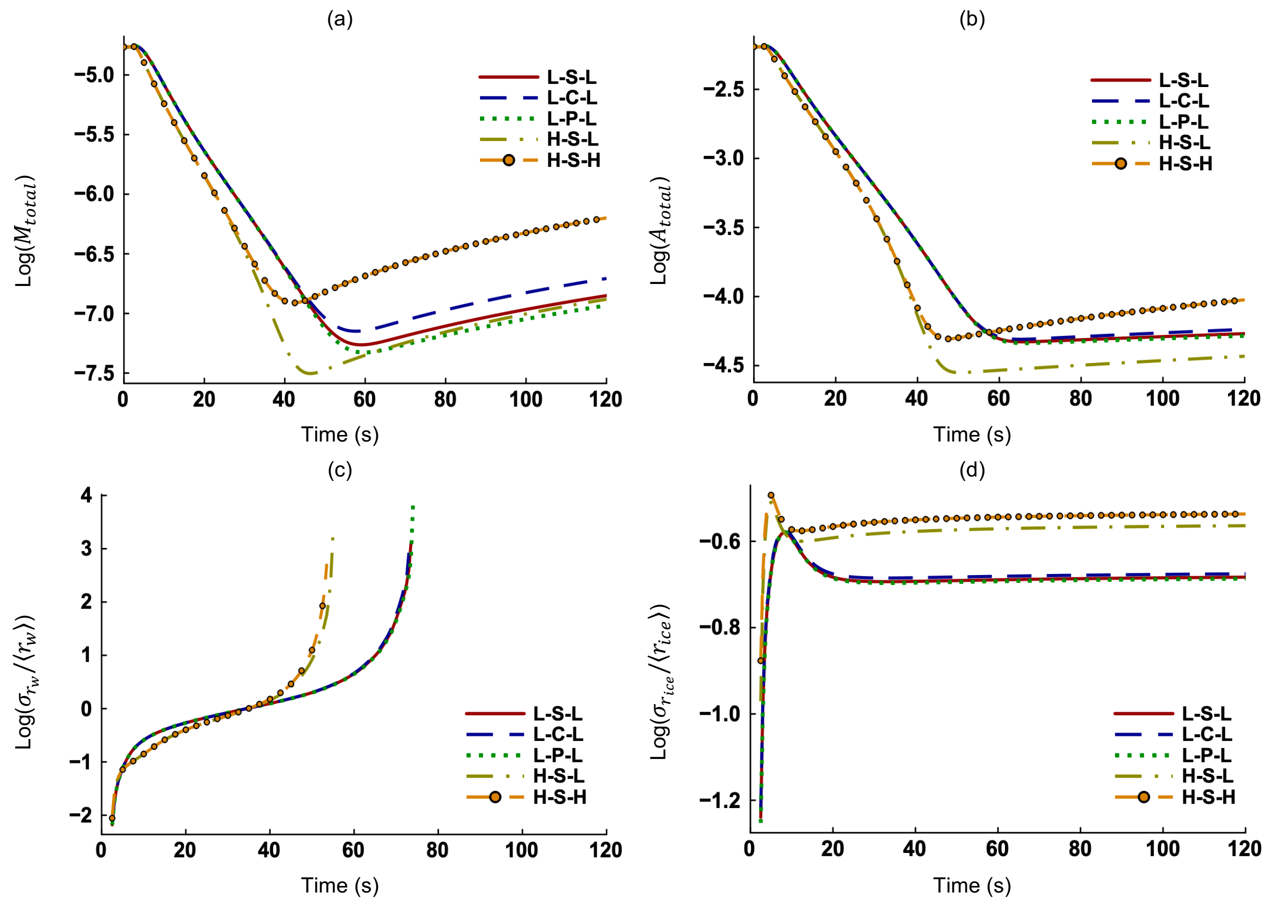


Figure 2. Time evolution of the (a) total cloud mass (M_{cloud}), (b) total cloud area (A_{cloud}), (c) relative dispersion of cloud droplets ($\sigma_{r_w}/\langle r_w\rangle$), and (d) relative dispersion of ice crystals ($\sigma_{r_{ice}}/\langle r_{ice}\rangle$). The abscissa is shown on a logarithmic scale of base 10.

The combined mass and area of cloud droplets and ice crystals decrease first (Figures 2(a) and 2(b)), corresponding to strong evaporation. They increase afterwards as depositional growth of ice continues and evaporation is complete. The increase of M_{total} and A_{total} is maximum if the turbulence level is high and the ice number density is high (Case H-S-H). For low turbulence cases, the order of increase is: L-C-L > L-S-L > L-P-L, consistent with Figures 1(c) and 1(d). The total cloud area increases least if the turbulence level is high and the ice number density is low (Case H-S-L). The albedo (reflectivity) of a cloud depends on both its mass and area. Hence, the statistics provided by our DNS model will be useful to determine or parameterize mixed-phase cloud albedo. The relative dispersion of the cloud droplet radii ($\sigma_{r_w}/\langle r_w \rangle$) increases with time (Figure 2(c)) and vanishes when cloud droplets no longer exist. It appears to shoot up just before evaporation ends for the respective cases. Extremely small cloud droplets evaporate instantaneously. The relative dispersion of the ice crystal radii ($\sigma_{r_{ice}}/\langle r_{ice} \rangle$) increases very fast initially when the monodisperse ice crystals are exposed to turbulent mixing (Figure 2(d)). The $\sigma_{r_{ice}}/\langle r_{ice} \rangle$ becomes statistically stationary with time, implying that the standard deviation $\sigma_{r_{ice}}$ and the mean radii $\langle r_{ice} \rangle$ increase equally. We also find that a low turbulence level (L-S-L, L-C-L, and L-P-L) has a retarding effect on the $\sigma_{r_{ice}}/\langle r_{ice} \rangle$ (Figure 2(d)).

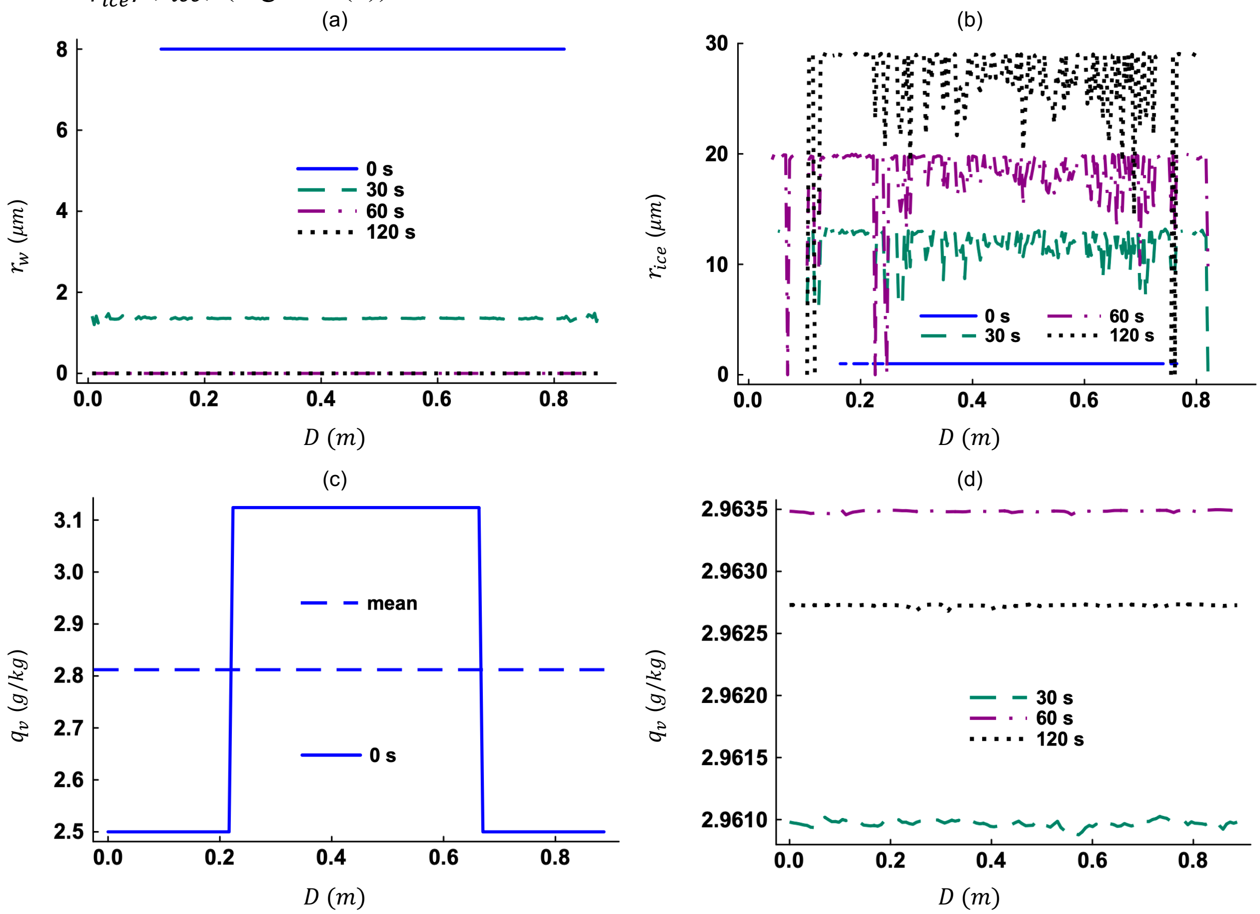


Figure 3. The spatial distributions of the (a) r_w and (b) r_{ice} at selected times. The spatial distributions of the q_v at (c) 0 s and at (d) other selected times. Note that D is the main diagonal of our 3D domain. These results represent Case H-S-L.

The distributions of the r_w and r_{ice} along the main diagonal (Figures 3(a)-(b)) suggest that ice crystal growth experiences more variation compared to that of cloud droplets. Because ice crystals are fewer in number and the domain is rich in water vapor, vapor molecules can freely deposit on ice crystal surfaces. This allows ice crystals to grow at different rates. In contrast, cloud droplets evaporate almost at the same rate (Figure 3(a)). Evaporation is completed by 55 s for Case

H-S-L (Figure 1(b)) and thus $r_w = 0$ at 60 s and 120 s in Figure 3(a). The spatial distribution of the q_v at the initial time shows that sharp interfaces are present (Figure 3(c)), which is consistent with Equation (15). Figure 3(d) suggests that the mean of q_v is more significant compared to the fluctuating component. Hence, phase change events like evaporation and deposition influence the water vapor mixing ratio more strongly than turbulent transport does. The q_v increases with time from a mean value of 2.812 g/kg at 0 s as evaporation releases water vapor in the domain. However, it decreases between 60 s and 120 s (Figure 3(d)) when only deposition takes place and water vapor is converted to solid ice.

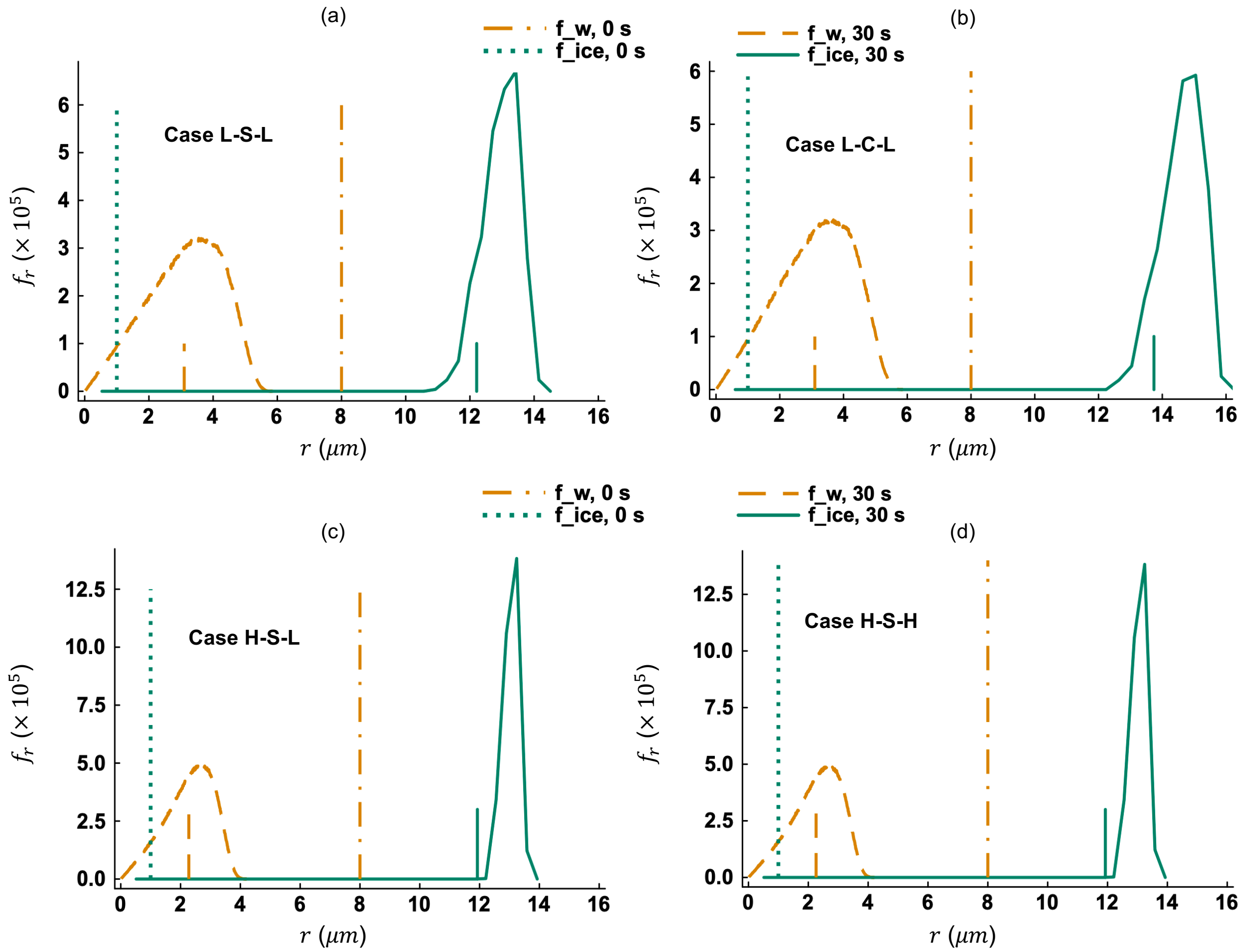


Figure 4. Probability density functions (PDF) of cloud droplet and ice crystal radii for Cases (a) L-S-L, (b) L-C-L, (c) H-S-L, and (d) H-S-H. The vertical line segments at the bottom indicate the expectation (mean) of r from respective PDFs.

The PDFs of cloud droplet radii (f_{r_w}) broadens for all cases between 0 s and 30 s (Figures 4(a)-(d)) because of turbulent mixing. The f_{r_w} is narrower at 30 s for high turbulence cases (Figures 4(c)-(d)) compared to that for low turbulence cases (Figures 4(a)-(b)). The narrowing is caused by a reduction in maximum cloud droplet radius. Hence, strong turbulence leads to partial evaporation of more cloud droplets. Broadening is also observed in the PDFs of ice crystal radii ($f_{r_{ice}}$) at 30 s relative to that at 0 s. A tiny fraction of ice crystals sublimate for all cases due to entrainment of dry air into the cloud volume. Therefore, the $f_{r_{ice}}$ has a long tail to the left of the mean. The peak of $f_{r_{ice}}$ is noticeably higher than that of f_{r_w}. The mean and maximum ice crystal radii are larger when the turbulence level is low and ice crystal shape is hexagonal column (Case L-C-L).

5 Concluding Remarks

The glaciation of a mixed-phase cloud, caused by entrainment and turbulent mixing, has been studied using a three-dimensional particle-based DNS model. Our findings clarify several microphysical aspects of the glaciation process. A high level of flow turbulence intensifies mixing between the cloud and environment and expedites glaciation. Glaciation is also enhanced if the ice number density is high. A hexagonal column geometry promotes ice growth and glaciation while a hexagonal plate shape does the opposite. The present letter shows that particle-based DNS can faithfully distinguish between cloud droplets and ice crystals, regardless of their sizes, at various stages of mixed-phase cloud growth.

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Open Research

The data that support the findings of this work are available from the corresponding author upon reasonable request.

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